

Robot data



DROID

real Franka arm episodes; split by recording session



Language-Table

xArm pushing blocks; long unscripted play data

Episode / session

one demonstration; episodes recorded consecutively at one site

End-effector

the tool at the arm's tip, a two-finger gripper

Action block

all commands of one model step, concatenated

ShiftWM terms



Patch features
frozen DINOv2
16x16 grid



Transport
content moved
from observed
patches



Gate
per-patch weight:
keep vs. move



Transport window
w×w patches, last
S frames



Correction
added residual
for new content



Oracle move
observed feature
closest to truth

Surgical data



Open-H Hamlyn

dVRK recordings of seven practice tasks



Practice task

a standard surgeon exercise, e.g. peg transfer



Phantom

an artificial practice model of tissue

dVRK

da Vinci Research Kit, surgical robot

Scene camera

external camera, not an endoscope

Simulated suites (plug-in, planning)



PushT

push a T-shaped block to a target pose



Wall

2-D agent moves through a door in a wall



TwoRoom

2-D navigation between two rooms (LeWM)



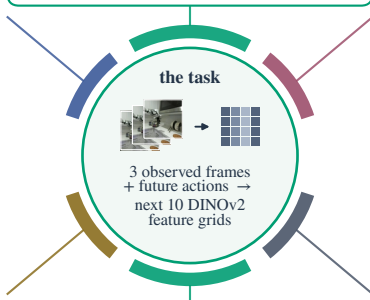
Reacher

a two-link arm reaches a target (LeWM)



MPC / CEM

plan, act, re-plan; sample and refit



Evaluation terms



Persistence
copy the last
observed grid



Moving patches
25% with largest
true change



Skill
share of
persistence error
removed



Rollout
own predictions
fed back as
inputs



Teacher forcing
train on true
past, not own
output



Session split
no recording
session in two
splits

World models and baselines

- **V-JEPA 2-AC**
published action-conditioned WM, post-trained on DROID
- **DINO-WM**
published WM on DINOv2 patch features
- ★ **LeWM**
released end-to-end WM; planning reference
- ◆ **Direct**
regresses each future grid:
 $Z_0 + \text{learned residual}$
- ▲ **AR**
residual to the latest grid, rolled out recursively
- **AR-TF**
DINO-WM-style: teacher-forced next step, rolled out